Aramark Gateway Interface Specification

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1.0 Introduction

This document describes the interface to USA Technologies' Aramark Gateway. The Aramark Gateway is a web service that handles requests for Aramark services from the Aramark Relay Transaction Server. The Aramark Gateway receives and processes SOAP messages that represent requests for Aramark transaction services. The Aramark Gateway returns a response, also in the form of a SOAP message, that represents the result of a request for Aramark transaction services.

There are five types of transaction services supported by the Aramark Gateway: inquiry, decliningBalance, credit, cashEquivalency and board. The cashEquivalency and board transaction services are currently not supported by the Aramark Relay Transaction Server.

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2.0 Transaction Services

Applications can request Aramark Transaction Services from the Aramark Gateway by sending a properly formatted SOAP message to the gateway. The gateway will accept such messages using HTTP as a transport protocol. The Aramark Gateway is implemented as an Apache Axis web service.

Any application that requests Aramark services through the gateway must provide several data elements as listed in the following sections. The request must pass validation tests as documented in the XML Schema in section 3.0. Please consult "ARAMARK Campus Services POS Interface" by Aramark for more information regarding the elements listed below.

	Element	Description	Required
1	remoteServerAddress	URL of Aramark Relay Server's web service	*
2	account	Account Name ** Must be same name as setup on Relay Transaction Server & relates to a POS device	*
3	cardNumber	Customer Account Number	*
4	resultDate	Date and Time of the Transaction	*
5	operatorID	Operation Number from system sending over request	
6	transactionID	ID of the Transaction	
7	revenueCenterID	Optional Field to record transactions to a Revenue Center	
8	offline	"TRUE" or "FALSE"	*
9	debitAmount	Amount to be deducted from the customer's account	* for declining Balance request
10	creditAmount	Amount to be credited to the customer's account	* for credit request
11	enableEncryption	"TRUE" of "FALSE"	*
12	cryptoKey	Encryption key to use when communicating with Aramark Relay Transaction Server. Each terminal may have its own encryption key.	* if encryption is enabled
13	cryptoIV	Encryption initialization vector to use when communicating with Aramark Relay Transaction Server. Each terminal may have its own encryption initialization vector.	* if encryption is enabled

The reply returned by the gateway follows a common pattern. Please consult "ARAMARK Campus Services POS Interface" by Aramark for more information regarding the elements listed below. Return values include:

POSResponse

	Element	Type	Description
1	POSResponse	struct	Root element of POSResponse
2	POSResponse_Transaction	struct	A single Transaction that's been responded to from a POS

POSResponse_Transaction

Element	Type	Description
POSResponse_Transaction	struct	Subtype of POSResponse
DateTimeStamp	string	DateTime stamp of a single Transaction that's been

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		responded to from a POS
TransactionTypeCode	string	INQUIRY, DECLININGBALANCE, CREDIT, CASHEQUIVALENCY, BOARD
Status	string	Transaction success code (Valid, InValid, UnKnown, NotRegistered)
StatusCode	string	POS Transaction response code
ResponseMessage	string	POS message, Potentially for Customer
ResponseText	string	POS Text / Error Messages
POSResponse_Customer	struct	Customer that this transaction relates to

POSResponse_Customer

Element	Type	Description
POSResponse_Customer	struct	Subtype of POSResponse_Transaction
CustomerID	string	Customer Card Number
CustomerIssueID	string	Customer Card Number Issue Number
FirstName	string	Customer First name
LastName	string	Customer Last name
Amount	string	Customer Account Balance
TaxLines	struct	Structure of TaxLines

TaxLines

Element	Type	Description
TaxAuthorityID	string	TaxNumber
UseTax	string	TRUE or FALSE value if tax is used

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2.1 Inquiry

The Inquiry method allows for inquires to be requested to the POS system. An XML response will be sent back with the detailed information below. Please see the XML schema definitions in section 3.0 for more information. Inquiry transactions are always online, so the Offline tag must be set to FALSE.

Below is a sample SOAP message that implements an inquiry request.

```
<?xml version="1.0" encoding="UTF-8"?>
<SOAP-ENV:Envelope xmlns:xsi="http://www.w3.org/1999/XMLSchema-instance" xmlns:SOAP-
ENC="http://schemas.xmlsoap.org/soap/encoding/" xmlns:SOAP-
ENV="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/1999/XMLSchema" SOAP-
ENV:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
  <SOAP-ENV:Body>
        <inquiry>
          <remoteServerAddress
          xsi:type="xsd:string">http://demo.aramarkcampusit.com/Relay_V2/RelayService.asmx
          </remoteServerAddress>
           <account xsi:type="xsd:string">ScanPlus_90</account>
           <cardNumber xsi:type="xsd:string">999999999</cardNumber>
           <resultDate xsi:type="xsd:string">2004-12-02T10:05:00.0000000-04:00</resultDate>
           <operatorID xsi:type="xsd:string"/>
           <transactionID xsi:type="xsd:string">123456</transactionID>
           <revenueCenterID xsi:type="xsd:string"/>
           <offline xsi:type="xsd:string">FALSE</offline>
           <enableEncryption xsi:type="xsd:string">TRUE</enableEncryption>
           <cryptoKey xsi:type="xsd:string">g9f9tr4j34t66lkjgovjo8rs</cryptoKey>
           <cryptoIV xsi:type="xsd:string">vgv8gfd9</cryptoIV>
        </inquiry>
  </SOAP-ENV:Body>
</SOAP-ENV:Envelope>
```

Below is a sample SOAP message that implements an inquiry response.

```
<?xml version="1.0" encoding="UTF-8"?>
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/1999/XMLSchema" xmlns:xsi="http://www.w3.org/1999/XMLSchema-
instance">
 <soapenv:Body>
  <inquiryResponse soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
   <inquiryReturn href="#id0"/>
  </inquiryResponse>
  <multiRef id="id0" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns1:POSResponse" xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:ns1="http://relayservice.webservices.educationaramark.com">
   <Account xsi:type="ns2:string"
xmlns:ns2="http://www.w3.org/2001/XMLSchema">ScanPlus_90</Account>
   <Transaction href="#id1"/>
  </multiRef>
  <multiRef id="id1" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns3:POSResponse_Transaction" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns:ns3="http://relayservice.webservices.educationaramark.com"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
   <BeginDateTimeStamp xsi:type="ns4:string"
xmlns:ns4="http://www.w3.org/2001/XMLSchema">12/16/2004 12:01:50 PM</BeginDateTimeStamp>
   <TransactionTypeCode xsi:type="ns5:string"
xmlns:ns5="http://www.w3.org/2001/XMLSchema">INQUIRY</TransactionTypeCode>
   <Status xsi:type="ns6:string"
xmlns:ns6="http://www.w3.org/2001/XMLSchema">VALID</Status>
   <StatusCode xsi:type="ns7:string"
xmlns:ns7="http://www.w3.org/2001/XMLSchema">0</StatusCode>
```

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```
   <ResponseMessage xsi:type="ns8:string"
xmlns:ns8="http://www.w3.org/2001/XMLSchema">Customer Balance is
$235.00</ResponseMessage>
   <ResponseText xsi:type="ns9:string" xmlns:ns9="http://www.w3.org/2001/XMLSchema">VALID
$235.00</ResponseText>
   <Customer href="#id2"/>
  </multiRef>
  <multiRef id="id2" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns10:POSResponse_Customer" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns:ns10="http://relayservice.webservices.educationaramark.com"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
   <CustomerID xsi:type="ns11:string"
xmlns:ns11="http://www.w3.org/2001/XMLSchema">999999999</CustomerID>
   <CusotmerIssueID xsi:type="ns12:string"
xmlns:ns12="http://www.w3.org/2001/XMLSchema"></CusotmerIssueID>
   <FirstName xsi:type="ns13:string"
xmlns:ns13="http://www.w3.org/2001/XMLSchema">Test1</FirstName>
   <LastName xsi:type="ns14:string"
xmlns:ns14="http://www.w3.org/2001/XMLSchema">Test1</LastName>
   <Amount xsi:type="ns15:string"
xmlns:ns15="http://www.w3.org/2001/XMLSchema">235.00</Amount>
   <Tax href="#id3"/>
  </multiRef>
  <multiRef id="id3" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns16:ArrayOfTaxLines" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:ns16="http://relayservice.webservices.educationaramark.com"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
   <TaxLines href="#id4"/>
   <TaxLines href="#id5"/>
   <TaxLines href="#id6"/>
   <TaxLines href="#id7"/>
  </multiRef>
  <multiRef id="id7" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns17:TaxLines" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:ns17="http://relayservice.webservices.educationaramark.com"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
   <TaxAuthorityID xsi:type="ns18:string"
xmlns:ns18="http://www.w3.org/2001/XMLSchema">Tax4</TaxAuthorityID>
   <UseTax xsi:type="ns19:string"
xmlns:ns19="http://www.w3.org/2001/XMLSchema">TRUE</UseTax>
  </multiRef>
  <multiRef id="id4" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns20:TaxLines" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:ns20="http://relayservice.webservices.educationaramark.com"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
   <TaxAuthorityID xsi:type="ns21:string"
xmlns:ns21="http://www.w3.org/2001/XMLSchema">Tax1</TaxAuthorityID>
   <UseTax xsi:type="ns22:string"
xmlns:ns22="http://www.w3.org/2001/XMLSchema">TRUE</UseTax>
  </multiRef>
  <multiRef id="id6" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns23:TaxLines" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:ns23="http://relayservice.webservices.educationaramark.com"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
   <TaxAuthorityID xsi:type="ns24:string"
xmlns:ns24="http://www.w3.org/2001/XMLSchema">Tax3</TaxAuthorityID>
   <UseTax xsi:type="ns25:string"
xmlns:ns25="http://www.w3.org/2001/XMLSchema">TRUE</UseTax>
  </multiRef>
  <multiRef id="id5" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns26:TaxLines" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:ns26="http://relayservice.webservices.educationaramark.com"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
   <TaxAuthorityID xsi:type="ns27:string"
xmlns:ns27="http://www.w3.org/2001/XMLSchema">Tax2</TaxAuthorityID>
```


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```
   <UseTax xsi:type="ns28:string"
xmlns:ns28="http://www.w3.org/2001/XMLSchema">TRUE</UseTax>
  </multiRef>
 </soapenv:Body>
</soapenv:Envelope>
```


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2.2 Declining Balance

The Declining Balance transaction method allows for debits to be deducted from a customer's account. An XML response will be sent back with the detailed information below. Please see the XML schema definitions in section 3.0 for more information.

Below is a sample SOAP message that implements a decliningBalance request.

```
<?xml version="1.0" encoding="UTF-8"?>
<SOAP-ENV:Envelope xmlns:xsi="http://www.w3.org/1999/XMLSchema-instance" xmlns:SOAP-
ENC="http://schemas.xmlsoap.org/soap/encoding/" xmlns:SOAP-
ENV="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/1999/XMLSchema" SOAP-
ENV:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
  <SOAP-ENV:Body>
    <decliningBalance>
        <remoteServerAddress
        xsi:type="xsd:string">http://demo.aramarkcampusit.com/Relay_V2/RelayService.asmx</
        remoteServerAddress>
        <account xsi:type="xsd:string">ScanPlus_90</account>
        <cardNumber xsi:type="xsd:string">999999999</cardNumber>
        <resultDate xsi:type="xsd:string">2004-12-02T10:05:00.0000000-04:00</resultDate>
        <operatorID xsi:type="xsd:string"/>
        <transactionID xsi:type="xsd:string">123456</transactionID>
        <revenueCenterID xsi:type="xsd:string"/>
        <offline xsi:type="xsd:string">FALSE</offline>
        <debitAmount xsi:type="xsd:string">1.00</debitAmount>
        <enableEncryption xsi:type="xsd:string">TRUE</enableEncryption>
        <cryptoKey xsi:type="xsd:string">g9f9tr4j34t66lkjgovjo8rs</cryptoKey>
        <cryptoIV xsi:type="xsd:string">vgv8gfd9</cryptoIV>
    </decliningBalance>
  </SOAP-ENV:Body>
</SOAP-ENV:Envelope>
```

Below is a sample SOAP message that implements a decliningBalance response.

```
<?xml version="1.0" encoding="UTF-8"?>
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/1999/XMLSchema" xmlns:xsi="http://www.w3.org/1999/XMLSchema-
instance">
 <soapenv:Body>
  <decliningBalanceResponse
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
   <decliningBalanceReturn href="#id0"/>
  </decliningBalanceResponse>
  <multiRef id="id0" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns1:POSResponse" xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:ns1="http://relayservice.webservices.educationaramark.com">
   <Account xsi:type="ns2:string"
xmlns:ns2="http://www.w3.org/2001/XMLSchema">ScanPlus_90</Account>
   <Transaction href="#id1"/>
  </multiRef>
  <multiRef id="id1" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns3:POSResponse_Transaction" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns:ns3="http://relayservice.webservices.educationaramark.com"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
   <BeginDateTimeStamp xsi:type="ns4:string"
xmlns:ns4="http://www.w3.org/2001/XMLSchema">12/16/2004 1:26:29 PM</BeginDateTimeStamp>
   <TransactionTypeCode xsi:type="ns5:string"
xmlns:ns5="http://www.w3.org/2001/XMLSchema">DECLININGBALANCE</TransactionTypeCode>
   <Status xsi:type="ns6:string"
xmlns:ns6="http://www.w3.org/2001/XMLSchema">VALID</Status>
   <StatusCode xsi:type="ns7:string"
xmlns:ns7="http://www.w3.org/2001/XMLSchema">0</StatusCode>
```

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```
   <ResponseMessage xsi:type="ns8:string"
xmlns:ns8="http://www.w3.org/2001/XMLSchema">Customer Balance is
$234.00</ResponseMessage>
   <ResponseText xsi:type="ns9:string" xmlns:ns9="http://www.w3.org/2001/XMLSchema">VALID
$234.00</ResponseText>
   <Customer href="#id2"/>
  </multiRef>
  <multiRef id="id2" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns10:POSResponse_Customer" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns:ns10="http://relayservice.webservices.educationaramark.com"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
   <CustomerID xsi:type="ns11:string"
xmlns:ns11="http://www.w3.org/2001/XMLSchema">999999999</CustomerID>
   <CusotmerIssueID xsi:type="ns12:string"
xmlns:ns12="http://www.w3.org/2001/XMLSchema"></CusotmerIssueID>
   <FirstName xsi:type="ns13:string"
xmlns:ns13="http://www.w3.org/2001/XMLSchema">Test1</FirstName>
   <LastName xsi:type="ns14:string"
xmlns:ns14="http://www.w3.org/2001/XMLSchema">Test1</LastName>
   <Amount xsi:type="ns15:string"
xmlns:ns15="http://www.w3.org/2001/XMLSchema">234.00</Amount>
   <Tax xsi:type="ns10:ArrayOfTaxLines" xsi:null="true"
xmlns:xsi="http://www.w3.org/1999/XMLSchema-instance"/>
  </multiRef>
 </soapenv:Body>
</soapenv:Envelope>
```


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2.3 Credit

The Credit transaction method allows for credits/refunds to be placed onto a customer's account. An XML response will be send back with the detailed information below.

Below is a sample SOAP message that implements a credit request.

```
<?xml version="1.0" encoding="UTF-8"?>
<SOAP-ENV:Envelope xmlns:xsi="http://www.w3.org/1999/XMLSchema-instance" xmlns:SOAP-
ENC="http://schemas.xmlsoap.org/soap/encoding/" xmlns:SOAP-
ENV="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/1999/XMLSchema" SOAP-
ENV:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
  <SOAP-ENV:Body>
    <credit>
        <remoteServerAddress
        xsi:type="xsd:string">http://demo.aramarkcampusit.com/Relay_V2/RelayService.asmx</
        remoteServerAddress>
       <account xsi:type="xsd:string">ScanPlus_90</account>
       <cardNumber xsi:type="xsd:string">999999999</cardNumber>
       <resultDate xsi:type="xsd:string">2004-12-02T10:05:00.0000000-04:00</resultDate>
       <operatorID xsi:type="xsd:string"/>
       <transactionID xsi:type="xsd:string">123456</transactionID>
       <revenueCenterID xsi:type="xsd:string"/>
       <offline xsi:type="xsd:string">FALSE</offline>
       <creditAmount xsi:type="xsd:string">1.00</creditAmount>
       <enableEncryption xsi:type="xsd:string">TRUE</enableEncryption>
       <cryptoKey xsi:type="xsd:string">g9f9tr4j34t66lkjgovjo8rs</cryptoKey>
       <cryptoIV xsi:type="xsd:string">vgv8gfd9</cryptoIV>
    </credit>
  </SOAP-ENV:Body>
</SOAP-ENV:Envelope>
```

Below is a sample SOAP message that implements a credit response.

```
<?xml version="1.0" encoding="UTF-8"?>
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/1999/XMLSchema" xmlns:xsi="http://www.w3.org/1999/XMLSchema-
instance">
 <soapenv:Body>
  <creditResponse soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
   <creditReturn href="#id0"/>
  </creditResponse>
  <multiRef id="id0" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns1:POSResponse" xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:ns1="http://relayservice.webservices.educationaramark.com">
   <Account xsi:type="ns2:string"
xmlns:ns2="http://www.w3.org/2001/XMLSchema">ScanPlus_90</Account>
   <Transaction href="#id1"/>
  </multiRef>
  <multiRef id="id1" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns3:POSResponse_Transaction" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns:ns3="http://relayservice.webservices.educationaramark.com"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
   <BeginDateTimeStamp xsi:type="ns4:string"
xmlns:ns4="http://www.w3.org/2001/XMLSchema">12/16/2004 1:36:52 PM</BeginDateTimeStamp>
   <TransactionTypeCode xsi:type="ns5:string"
xmlns:ns5="http://www.w3.org/2001/XMLSchema">CREDIT</TransactionTypeCode>
   <Status xsi:type="ns6:string"
xmlns:ns6="http://www.w3.org/2001/XMLSchema">VALID</Status>
   <StatusCode xsi:type="ns7:string"
xmlns:ns7="http://www.w3.org/2001/XMLSchema">0</StatusCode>
```

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```
   <ResponseMessage xsi:type="ns8:string"
xmlns:ns8="http://www.w3.org/2001/XMLSchema">Customer Balance is
$235.00</ResponseMessage>
   <ResponseText xsi:type="ns9:string" xmlns:ns9="http://www.w3.org/2001/XMLSchema">VALID
$235.00</ResponseText>
   <Customer href="#id2"/>
  </multiRef>
  <multiRef id="id2" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns10:POSResponse_Customer" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns:ns10="http://relayservice.webservices.educationaramark.com"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/">
   <CustomerID xsi:type="ns11:string"
xmlns:ns11="http://www.w3.org/2001/XMLSchema">999999999</CustomerID>
   <CusotmerIssueID xsi:type="ns12:string"
xmlns:ns12="http://www.w3.org/2001/XMLSchema"></CusotmerIssueID>
   <FirstName xsi:type="ns13:string"
xmlns:ns13="http://www.w3.org/2001/XMLSchema">Test1</FirstName>
   <LastName xsi:type="ns14:string"
xmlns:ns14="http://www.w3.org/2001/XMLSchema">Test1</LastName>
   <Amount xsi:type="ns15:string"
xmlns:ns15="http://www.w3.org/2001/XMLSchema">235.00</Amount>
   <Tax xsi:type="ns10:ArrayOfTaxLines" xsi:null="true"
xmlns:xsi="http://www.w3.org/1999/XMLSchema-instance"/>
  </multiRef>
 </soapenv:Body>
</soapenv:Envelope>
```


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3.0 WSDL / XML Schema

This section documents the Aramark Gateway web service description and XML schema used to validate requests. A complete listing of the gateway's WSDL file follows.


```xml
<?xml version="1.0" encoding="UTF-8"?>
        <wsdl:definitions targetNamespace="urn:aramark.usatech.com"
        xmlns="http://schemas.xmlsoap.org/wsdl/"
        xmlns:apachesoap="http://xml.apache.org/xml-soap"
        xmlns:impl="urn:aramark.usatech.com" xmlns:intf="urn:aramark.usatech.com"
        xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/"
        xmlns:tns1="http://relayservice.webservices.educationaramark.com"
        xmlns:wsdl="http://schemas.xmlsoap.org/wsdl/"
        xmlns:wsdlsoap="http://schemas.xmlsoap.org/wsdl/soap/"
        xmlns:xsd="http://www.w3.org/2001/XMLSchema"><wsdl:types><schema
        targetNamespace="http://relayservice.webservices.educationaramark.com"
        xmlns="http://www.w3.org/2001/XMLSchema"><import
        namespace="http://schemas.xmlsoap.org/soap/encoding/"/><complexType
        name="TaxLines"><sequence><element maxOccurs="1" minOccurs="0"
        name="TaxAuthorityID" nillable="true" type="xsd:string"/><element maxOccurs="1"
        minOccurs="0" name="UseTax" nillable="true"
        type="xsd:string"/></sequence></complexType><complexType
        name="ArrayOfTaxLines"><sequence><element maxOccurs="unbounded" minOccurs="0"
        name="TaxLines" nillable="true"
        type="tns1:TaxLines"/></sequence></complexType><complexType
        name="POSResponse_Customer"><sequence><element maxOccurs="1" minOccurs="0"
        name="CustomerID" nillable="true" type="xsd:string"/><element maxOccurs="1"
        minOccurs="0" name="CusotmerIssueID" nillable="true" type="xsd:string"/><element
        maxOccurs="1" minOccurs="0" name="FirstName" nillable="true"
        type="xsd:string"/><element maxOccurs="1" minOccurs="0" name="LastName"
        nillable="true" type="xsd:string"/><element maxOccurs="1" minOccurs="0"
        name="Amount" nillable="true" type="xsd:string"/><element maxOccurs="1"
        minOccurs="0" name="Tax" nillable="true"
        type="tns1:ArrayOfTaxLines"/></sequence></complexType><complexType
        name="POSResponse_Transaction"><sequence><element maxOccurs="1" minOccurs="0"
        name="BeginDateTimeStamp" nillable="true" type="xsd:string"/><element
        maxOccurs="1" minOccurs="0" name="TransactionTypeCode" nillable="true"
        type="xsd:string"/><element maxOccurs="1" minOccurs="0" name="Status"
        nillable="true" type="xsd:string"/><element maxOccurs="1" minOccurs="0"
        name="StatusCode" nillable="true" type="xsd:string"/><element maxOccurs="1"
        minOccurs="0" name="ResponseMessage" nillable="true" type="xsd:string"/><element
        maxOccurs="1" minOccurs="0" name="ResponseText" nillable="true"
        type="xsd:string"/><element name="Customer" nillable="true"
        type="tns1:POSResponse_Customer"/></sequence></complexType><complexType
        name="POSResponse"><sequence><element maxOccurs="1" minOccurs="0" name="Account"
        nillable="true" type="xsd:string"/><element name="Transaction" nillable="true"
        type="tns1:POSResponse_Transaction"/></sequence></complexType></schema><schema
        targetNamespace="urn:aramark.usatech.com"
        xmlns="http://www.w3.org/2001/XMLSchema"><import
        namespace="http://schemas.xmlsoap.org/soap/encoding/"/><complexType
        name="AramarkException"><sequence><element name="description" nillable="true"
        type="xsd:string"/><element name="responseMessage" nillable="true"
        type="xsd:string"/><element name="responseText" nillable="true"
        type="xsd:string"/><element name="status" nillable="true"
        type="xsd:string"/><element name="statusCode" nillable="true"
        type="xsd:string"/></sequence></complexType></schema></wsdl:types>
  <wsdl:message name="inquiryResponse">
    <wsdl:part name="inquiryReturn" type="tns1:POSResponse"/>
  </wsdl:message>
  <wsdl:message name="creditResponse">
    <wsdl:part name="creditReturn" type="tns1:POSResponse"/>
  </wsdl:message>
  <wsdl:message name="creditRequest">
    <wsdl:part name="remoteServerAddress" type="xsd:string"/>
    <wsdl:part name="account" type="xsd:string"/>
    <wsdl:part name="cardNumber" type="xsd:string"/>
    <wsdl:part name="resultDate" type="xsd:string"/>
    <wsdl:part name="operatorID" type="xsd:string"/>
```


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```
  <wsdl:part name="transactionID" type="xsd:string"/>
  <wsdl:part name="revenueCenterID" type="xsd:string"/>
  <wsdl:part name="offline" type="xsd:string"/>
  <wsdl:part name="creditAmount" type="xsd:string"/>
  <wsdl:part name="enableEncryption" type="xsd:string"/>
  <wsdl:part name="cryptoKey" type="xsd:string"/>
  <wsdl:part name="cryptoIV" type="xsd:string"/>
</wsdl:message>
<wsdl:message name="boardRequest">
  <wsdl:part name="remoteServerAddress" type="xsd:string"/>
  <wsdl:part name="account" type="xsd:string"/>
  <wsdl:part name="cardNumber" type="xsd:string"/>
  <wsdl:part name="resultDate" type="xsd:string"/>
  <wsdl:part name="operatorID" type="xsd:string"/>
  <wsdl:part name="transactionID" type="xsd:string"/>
  <wsdl:part name="revenueCenterID" type="xsd:string"/>
  <wsdl:part name="offline" type="xsd:string"/>
  <wsdl:part name="unitCount" type="xsd:string"/>
  <wsdl:part name="enableEncryption" type="xsd:string"/>
  <wsdl:part name="cryptoKey" type="xsd:string"/>
  <wsdl:part name="cryptoIV" type="xsd:string"/>
</wsdl:message>
<wsdl:message name="cashEquivalencyResponse">
  <wsdl:part name="cashEquivalencyReturn" type="tns1:POSResponse"/>
</wsdl:message>
<wsdl:message name="cashEquivalencyRequest">
  <wsdl:part name="remoteServerAddress" type="xsd:string"/>
  <wsdl:part name="account" type="xsd:string"/>
  <wsdl:part name="cardNumber" type="xsd:string"/>
  <wsdl:part name="resultDate" type="xsd:string"/>
  <wsdl:part name="operatorID" type="xsd:string"/>
  <wsdl:part name="transactionID" type="xsd:string"/>
  <wsdl:part name="revenueCenterID" type="xsd:string"/>
  <wsdl:part name="offline" type="xsd:string"/>
  <wsdl:part name="cashAmount" type="xsd:string"/>
  <wsdl:part name="enableEncryption" type="xsd:string"/>
  <wsdl:part name="cryptoKey" type="xsd:string"/>
  <wsdl:part name="cryptoIV" type="xsd:string"/>
</wsdl:message>
<wsdl:message name="inquiryRequest">
  <wsdl:part name="remoteServerAddress" type="xsd:string"/>
  <wsdl:part name="account" type="xsd:string"/>
  <wsdl:part name="cardNumber" type="xsd:string"/>
  <wsdl:part name="resultDate" type="xsd:string"/>
  <wsdl:part name="operatorID" type="xsd:string"/>
  <wsdl:part name="transactionID" type="xsd:string"/>
  <wsdl:part name="revenueCenterID" type="xsd:string"/>
  <wsdl:part name="offline" type="xsd:string"/>
  <wsdl:part name="enableEncryption" type="xsd:string"/>
  <wsdl:part name="cryptoKey" type="xsd:string"/>
  <wsdl:part name="cryptoIV" type="xsd:string"/>
</wsdl:message>
<wsdl:message name="getVersionRequest">
</wsdl:message>
<wsdl:message name="sampleMessageRequest">
</wsdl:message>
<wsdl:message name="declineingBalanceResponse">
  <wsdl:part name="declineingBalanceReturn" type="tns1:POSResponse"/>
</wsdl:message>
<wsdl:message name="getVersionResponse">
  <wsdl:part name="getVersionReturn" type="xsd:string"/>
</wsdl:message>
<wsdl:message name="boardResponse">
  <wsdl:part name="boardReturn" type="tns1:POSResponse"/>
</wsdl:message>
<wsdl:message name="AramarkException">
  <wsdl:part name="fault" type="impl:AramarkException"/>
</wsdl:message>
<wsdl:message name="declineingBalanceRequest">
  <wsdl:part name="remoteServerAddress" type="xsd:string"/>
  <wsdl:part name="account" type="xsd:string"/>
```

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```
    <wsdl:part name="cardNumber" type="xsd:string"/>
    <wsdl:part name="resultDate" type="xsd:string"/>
    <wsdl:part name="operatorID" type="xsd:string"/>
    <wsdl:part name="transactionID" type="xsd:string"/>
    <wsdl:part name="revenueCenterID" type="xsd:string"/>
    <wsdl:part name="offline" type="xsd:string"/>
    <wsdl:part name="debitAmount" type="xsd:string"/>
    <wsdl:part name="enableEncryption" type="xsd:string"/>
    <wsdl:part name="cryptoKey" type="xsd:string"/>
    <wsdl:part name="cryptoIV" type="xsd:string"/>
  </wsdl:message>
  <wsdl:message name="sampleMessageResponse">
    <wsdl:part name="sampleMessageReturn" type="tns1:POSResponse"/>
  </wsdl:message>
  <wsdl:portType name="WSClient">
    <wsdl:operation name="getVersion">
      <wsdl:input message="impl:getVersionRequest" name="getVersionRequest"/>
      <wsdl:output message="impl:getVersionResponse" name="getVersionResponse"/>
    </wsdl:operation>
    <wsdl:operation name="inquiry" parameterOrder="remoteServerAddress account cardNumber
resultDate operatorID transactionID revenueCenterID offline enableEncryption cryptoKey
cryptoIV">
      <wsdl:input message="impl:inquiryRequest" name="inquiryRequest"/>
      <wsdl:output message="impl:inquiryResponse" name="inquiryResponse"/>
      <wsdl:fault message="impl:AramarkException" name="AramarkException"/>
    </wsdl:operation>
    <wsdl:operation name="decliningBalance" parameterOrder="remoteServerAddress account
cardNumber resultDate operatorID transactionID revenueCenterID offline debitAmount
enableEncryption cryptoKey cryptoIV">
      <wsdl:input message="impl:decliningBalanceRequest" name="decliningBalanceRequest"/>
      <wsdl:output message="impl:decliningBalanceResponse"
name="decliningBalanceResponse"/>
      <wsdl:fault message="impl:AramarkException" name="AramarkException"/>
    </wsdl:operation>
    <wsdl:operation name="credit" parameterOrder="remoteServerAddress account cardNumber
resultDate operatorID transactionID revenueCenterID offline creditAmount enableEncryption
cryptoKey cryptoIV">
      <wsdl:input message="impl:creditRequest" name="creditRequest"/>
      <wsdl:output message="impl:creditResponse" name="creditResponse"/>
      <wsdl:fault message="impl:AramarkException" name="AramarkException"/>
    </wsdl:operation>
    <wsdl:operation name="board" parameterOrder="remoteServerAddress account cardNumber
resultDate operatorID transactionID revenueCenterID offline unitCount enableEncryption
cryptoKey cryptoIV">
      <wsdl:input message="impl:boardRequest" name="boardRequest"/>
      <wsdl:output message="impl:boardResponse" name="boardResponse"/>
      <wsdl:fault message="impl:AramarkException" name="AramarkException"/>
    </wsdl:operation>
    <wsdl:operation name="cashEquivalency" parameterOrder="remoteServerAddress account
cardNumber resultDate operatorID transactionID revenueCenterID offline cashAmount
enableEncryption cryptoKey cryptoIV">
      <wsdl:input message="impl:cashEquivalencyRequest" name="cashEquivalencyRequest"/>
      <wsdl:output message="impl:cashEquivalencyResponse"
name="cashEquivalencyResponse"/>
      <wsdl:fault message="impl:AramarkException" name="AramarkException"/>
    </wsdl:operation>
    <wsdl:operation name="sampleMessage">
      <wsdl:input message="impl:sampleMessageRequest" name="sampleMessageRequest"/>
      <wsdl:output message="impl:sampleMessageResponse" name="sampleMessageResponse"/>
      <wsdl:fault message="impl:AramarkException" name="AramarkException"/>
    </wsdl:operation>
  </wsdl:portType>
  <wsdl:binding name="aramarkgatewaySoapBinding" type="impl:WSClient">
    <wsdlsoap:binding style="rpc" transport="http://schemas.xmlsoap.org/soap/http"/>
    <wsdl:operation name="getVersion">
      <wsdlsoap:operation soapAction=""/>
      <wsdl:input name="getVersionRequest">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:input>
      <wsdl:output name="getVersionResponse">
```

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```
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:output>
    </wsdl:operation>
    <wsdl:operation name="inquiry">
      <wsdlsoap:operation soapAction=""/>
      <wsdl:input name="inquiryRequest">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:input>
      <wsdl:output name="inquiryResponse">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:output>
      <wsdl:fault name="AramarkException">
        <wsdlsoap:fault encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:fault>
    </wsdl:operation>
    <wsdl:operation name="decliningBalance">
      <wsdlsoap:operation soapAction=""/>
      <wsdl:input name="decliningBalanceRequest">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:input>
      <wsdl:output name="decliningBalanceResponse">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:output>
      <wsdl:fault name="AramarkException">
        <wsdlsoap:fault encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:fault>
    </wsdl:operation>
    <wsdl:operation name="credit">
      <wsdlsoap:operation soapAction=""/>
      <wsdl:input name="creditRequest">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:input>
      <wsdl:output name="creditResponse">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:output>
      <wsdl:fault name="AramarkException">
        <wsdlsoap:fault encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:fault>
    </wsdl:operation>
    <wsdl:operation name="board">
      <wsdlsoap:operation soapAction=""/>
      <wsdl:input name="boardRequest">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:input>
      <wsdl:output name="boardResponse">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:output>
      <wsdl:fault name="AramarkException">
        <wsdlsoap:fault encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:fault>
    </wsdl:operation>
    <wsdl:operation name="cashEquivalency">
      <wsdlsoap:operation soapAction=""/>
      <wsdl:input name="cashEquivalencyRequest">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:input>
      <wsdl:output name="cashEquivalencyResponse">
```

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```
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:output>
      <wsdl:fault name="AramarkException">
        <wsdlsoap:fault encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:fault>
    </wsdl:operation>
    <wsdl:operation name="sampleMessage">
      <wsdlsoap:operation soapAction=""/>
      <wsdl:input name="sampleMessageRequest">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:input>
      <wsdl:output name="sampleMessageResponse">
        <wsdlsoap:body encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:output>
      <wsdl:fault name="AramarkException">
        <wsdlsoap:fault encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
namespace="urn:aramark.usatech.com" use="encoded"/>
      </wsdl:fault>
    </wsdl:operation>
  </wsdl:binding>
  <wsdl:service name="WSClientService">
    <wsdl:port binding="impl:aramarkgatewaySoapBinding" name="aramarkgateway">
      <wsdlsoap:address location="http://localhost:9100/axis/services/aramarkgateway"/>
    </wsdl:port>
  </wsdl:service>
</wsdl:definitions>
```

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4.0 Transport Protocol

This section documents the Aramark Gateway transport protocol. A request for Aramark services will contain a header followed by the SOAP request.

Below is a sample header produced by Perl SOAP::Lite client that implements an inquiry request.

```
POST /axis/services/aramarkgateway HTTP/1.1
TE: deflate,gzip;q=0.3
Connection: TE, close
Accept: text/xml
Accept: multipart/*
Host: localhost:12345
User-Agent: SOAP::Lite/Perl/0.60
Content-Length: 962
Content-Type: text/xml; charset=utf-8
SOAPAction: "#inquiry"
```

The reply from the Aramark Gateway will contain the following header followed by the SOAP response.

```
HTTP/1.1 200 OK
Set-Cookie: JSESSIONID=DF768ADA5AA5E2B688A2A8930E88BC4E; Path=/axis
Content-Type: text/xml;charset=utf-8
Date: Thu, 16 Dec 2004 17:02:00 GMT
Server: Apache-Coyote/1.1
Connection: close
```

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5.0 Error Handling

This section documents the error responses generated as a result of receiving exceptions or faulty messages. In such cases Aramark Gateway will return a message of AramarkException complex type in the SOAP response. If Aramark Gateway receives an invalid response from Aramark Relay Transaction Server, the responseMessage, responseText, status and statusCode fields will be populated with information received from Aramark Relay Transaction Server. In case of a general exception, the description field will contain the exception message.

Below is a sample AramarkException SOAP response received from Aramark Gateway after sending invalid customer information in the request.


```
<?xml version="1.0" encoding="UTF-8"?>
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance">
 <soapenv:Body>
  <soapenv:Fault>
   <faultcode>soapenv:Server.generalException</faultcode>
   <faultstring></faultstring>
   <detail>
    <ns1:fault href="#id0" xmlns:ns1="urn:aramark.usatech.com"/>
    <ns2:exceptionName
xmlns:ns2="http://xml.apache.org/axis/">com.usatech.aramark.webservice.AramarkException</
ns2:exceptionName>
   </detail>
  </soapenv:Fault>
  <multiRef id="id0" soapenc:root="0"
soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"
xsi:type="ns3:AramarkException" xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/"
xmlns:ns3="urn:aramark.usatech.com">
   <description xsi:type="xsd:string">Received INVALID response status from Aramark Web
Service</description>
   <responseMessage xsi:type="xsd:string">Customer not Registered</responseMessage>
   <responseText xsi:type="xsd:string">NOT REGISTERED</responseText>
   <status xsi:type="xsd:string">InValid</status>
   <statusCode xsi:type="xsd:string">-1</statusCode>
  </multiRef>
 </soapenv:Body>
</soapenv:Envelope>
```


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Campus Services

Information Technology

Declining/Inclining Balance POS Interface Requirements

Version 1.0

Authors:	Campus IT/Campus Marketing
Creation Date:	April 12, 2004
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Control Number:	
Version:	1.0

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1.0 Overview

1.1 Background

Introduction:

- This document contains the requirements for interfacing Declining/Inclining balance, and Inquiries into the ARAMARK POS Interface.
- This interface specifications are provided with this document. Together they will provide the information necessary to complete an interface to the ARAMARK POS systems utilized by many of ARAMARK's Clients.
- This interface will be designed to work over the standard TCP/IP Port 80.
- XML will be used as the format for data streams going between source systems, and the POS relay systems.
- To the extent possible, standard XML schemas will be utilized
- Triple DES will be used to Encrypt the data.

1.2 Current State

Brief:

- Campus Services utilizes several major POS systems. Two of these systems are operated by and mainted by Campus IT. These applications all utilize their own proprietary communications protocols by which external applications can interface with them. This document, and the attached interface specification is meant to bridge the interoperability gaps of the two POS systems, and reduce the complexity of integrating with ARAMARK systems

1.3 Objectives

Brief:

- Provide a common standards based approach to interfacing external systems with the POS systems utilized in the ARAMARK Campus Services environment.
- Provide flexibility and lower startup costs through the use of a well defined interface.
- Enable a secure process via the use of commonly available security tools.

1.4 Assumptions

Brief:

- Communications with the ARAMARK POS Interface can will go over the standard port: 80

1.6 Approach

Brief:

- Utilize a standards based framework. This includes the use of web services, and transactions based on the ARTS standards.
- Test via ARAMARK test server in Philadelphia

1.7 Scope

In Scope:

- ARAMARK has developed web services to support the following transaction types.
    - Inquiry
    - Inclining/Declining Balance
- The following transactions also exist, but are not a part of this implementation.
    - Cash Equivalency
    - Cash/Credit
    - Board Meals
    - Transaction Summary consisting of transaction totals, tenders, item detail, discounts, etc….

2.0 High-Level Functional Requirements

2.1 Overview

ARAMARK has developed specifications for a web services based interface that will be utilized as a hub to broker transactions between external sources and the standard POS system utilized on campus dining facilities managed by ARAMARK.

This interface will be developed as a Secure web service in Visual Studio .Net 2003.

The interface will utilize a basic framework for POS transactions developed by ARTS, the Association for Retail Technology Standards. The ARTS home page is located at http://www.nrf-arts.org . ARTS has developed a set of XML schemas to standardize communications between systems that handle POS data. These schemas cover areas from Tenders, to item detail, to discounts, and taxes.

ARAMARK has developed some extensions to that framework to handle University specific transactions like Declining Balance, Board Plans, and Cash Equivalency transactions.

The details of the interfaces for all transactions can be located in the ARAMARK Campus Services POS Interface document.

2.2 System Configuration

2.2.1 Configuration

Every call to the web service must have a client and device identifier. This is a unique value which identifies to the web service what school, and device the transaction is originating from. These will be mapped to appropriate devices in the pos system the transaction is being sent to.

In addition, the PC originating the request will have to know the IP address, or DNS name of web service.

2.3 Supported Transactions

2.3.2 Inquiry

Inquiry Transactions are utilized to request the balance a patron has on the POS system without actually modifying any funds. The request sent to the POS system includes the card number of a patron. The POS interface will reply with the balance available to the patron. If the patron has a balance available, they can spend up to that amount. It is possible for a patron to spend beyond the balance on the card. If this is the case, the system will show that the user has a credit balance.

- An inquiry should be performed as the first step in performing a declining balance which will be described next.
- Inquiries may also be performed in a standalone manner to determine the available balance of the patron

2.3.2 Declining Balance

Declining Balance involves the use of available funds from an account on one of the ARAMARK POS systems. Declining Balance is generally a two step process Part one of the process is to perform an inquiry transaction as described in section 2.3.1. The system will return with the available balance for the patron. If the patron has sufficient funds, a declining balance transaction is then sent to remove those funds from the patron's account.

- If the student does not have sufficient funds to cover the whole transaction, then the portion of the transaction indicated by the balance returned from the inquiry should be sent as the declining balance amount. The originating system should then prompt the user for another payment for the remaining balance on the transaction
- While the patron has funds available, the inquiry will indicate this amount. If the patron has 0, or less money, then the system will indicate if the student can continue to spend money in a credit mode.
- Some schools work exclusively in a credit or inclining balance. This is done with payroll deduction as well, where we create a monthly statement on the POS system. In this case, the POS system will always indicate that the patron has a credit balance.
- The use of declining balance generally has tax implications. The Inquiry transaction to the POS system will return a tax flag indicating whether items charged to that DB account should be taxed or not. If items were taxable, but the flag indicates not to use taxes, then the tax amount should be removed prior to processing the transaction.

2.3.3 Credit to Account

Credit to Account is not a credit card transaction. Credit to Account is used add money back to a Declining Balance account.

- The ARAMARK interface will have a configuration to determine whether Credit Account Transactions are allowed for a given device.

2.5 Operator Procedures

2.5.1 Overview

Appropriate procedures should exist in the host system for managing processes required by the addition of the communication mechanism, and transaction processing described in this document. This includes configuring the register with data required by the POS Interface,

2.5.2 Configuration

Ideally, there should be a place within the host software to configure settings required for communications with the ARAMARK web service.

2.5.3 Reporting

Reporting should be available in the host system to reconcile inclining and declining balance transactions against the ARAMARK POS system. An equivalent report would run on the ARAMARK POS system to detail the same transactions for the same time period. Any discrepancies should be researched and addressed.

3.0 Appendix

3.1 POS Interface Specifications

See attached documents

3.2 POS Interface Error Codes

See attached documents

ARAMARK Campus Services POS Interface

XML interfaces

ARAMARK Confidential

Prepared by
Campus Services IT

May 25, 2004

Relay Transaction Server: Response Messages

Attached are the ARAMARK Campus Services Replay Transaction Server response messages and status codes. These messages and status codes will be standardized from multiple POS Systems the Transaction Server will interact between.

The Status tag message will be Valid, InValid and UnKnown. These messages will represent if the transaction was processed successfully. The StatusCode number will represent the actual success issue or error of the Status tag. The ResponseMessage tag was created to be a response message back to the customer. This message can be customized to your specifications if needed. The ResponseText is primarily used for Error messages and for a subset of the response back from the POS system.

Status Codes

0+		Successful Return values
-1	.. –1000	Processor Messages
-1001	.. –2000	Transaction Messages
-2001	.. –3000	Communication Responses

Status	StatusCode	Definition
Valid	0	ResponseText: VALID $XXXX.XX ResponseMessage: Customer Balance is XXXXX.XX
Valid	100	ResponseText: CR BAL $XX.XX ResponseMessage: Customer Balance is -XXXXX.XX

Status	StatusCode	Definition
InValid	-1	ResponseText: NOT REGISTERED ResponseMessage: Customer not Registered
InValid	-2	ResponseText: EXPIRED CARD ResponseMessage: Expired Card
InValid	-3	ResponseText: LOST CARD,GRAB ResponseMessage: Card Reported Lost.
InValid	-4	ResponseText: SUSPENDED CARD ResponseMessage: Suspended Card

InValid	-5	ResponseText: SEE MANAGER ResponseMessage: See Manager
InValid	-6	ResponseText: OVER DAILY LIMIT ResponseMessage: Customer Over Daily Limit
InValid	-7	ResponseText: OLD ISSUE # ResponseMessage: Customer using invalid issued Card.
InValid	-8	ResponseText: INVALID ResponseMessage: Invalid Transaction
InValid	-9	**ResponseText:** FAILED XXXX **ResponseMessage:** FAILED XXXX
InValid	-10	ResponseText: INVALID FUNCTION ResponseMessage: Transaction can not be processed at this time.
InValid	-11	ResponseText: HALL CLOSED ResponseMessage: HALL CLOSED
InValid	-12	ResponseText: COMM ERROR-RETRY ResponseMessage: COMM ERROR-RETRY
InValid	-13	ResponseText: INVALID ZONE ResponseMessage: Transaction can not be completed during this time period.
InValid	-14	ResponseText: NOT CR MERCHANT ResponseMessage: NOT CR MERCHANT
InValid	-15	ResponseText: MEAL PL NOT FND ResponseMessage: Meal Plan assigned to patron does not exist.
InValid	-100	ResponseText: OFFLINE ResponseMessage: Offline

Transaction Specification Errors

Status	StatusCode	Definition
InValid	-1001	ResponseMessage: Begin DateTime could not be converted to a Date/Time.
InValid	-1002	ResponseMessage: End DateTime could not be converted to a Date/Time.
InValid	-1003	ResponseMessage: Offline entry was not a 'TRUE' or 'FALSE'.
InValid	-1004	ResponseMessage: Transaction Number formatting is incorrect.
InValid	-1005	ResponseMessage: Revenue Center Number formatting is incorrect.
InValid	-1006	ResponseMessage: Operator Formatting is incorrect.
InValid	-1007	ResponseMessage: Customer Number must be greater then 0 characters.
InValid	-1008	ResponseMessage: Amount formatting is incorrect.
InValid	-1009	ResponseMessage: Unit Count formatting is incorrect.
InValid	-1010	ResponseMessage: Invalid Transaction Type Code.
InValid	-1011	ResponseMessage: Amount is Greater/Less than Maximum Amount allow.
InValid	-1012	ResponseMessage: Amount can not be negative.
InValid	-1013	ResponseMessage: Transaction Number can not be greater then 6 Characters.
InValid	-1014	ResponseMessage: Transaction could not be Validated Successfully.
InValid	-1015	ResponseMessage: Incorrect Account Name parameter value Or Configuration settings not found for specified Account Name Or 'Invalid Account Name : Config File.
InValid	-1016	ResponseMessage: InValid POS Solution

Communication Specification Errors

Status	StatusCode	Definition
InValid	-2001	ResponseMessage: Server Error Message * Transaction could not be processed for an Unknown Server processing Issue
InValid	-2002	ResponseMessage: Server result did not parse correctly.

InValid	-2003	ResponseMessage: `Server not available using IP XXXXXXXX Port XXXX`

ARAMARK Campus Services POS Interface

XML interfaces

ARAMARK Confidential

Prepared by
Campus Services IT

April 1, 2026

Overview

Campus Services IT has developed an open interface standard for interaction between third party POS devices and POS systems. This open interface uses Web Services and XML to transfer POS transactions between the POS application/device and the Campus Services Relay Transaction Server. The Relay Transaction Server then redirects the transmission to the appropriate POS system for processing and responds back to the application/device with the POS response. The benefit of using this open interface is that communication to multiple POS systems can be made using this one interface specification. This document outlines the interfaces that are available and the security settings.

Security Overview

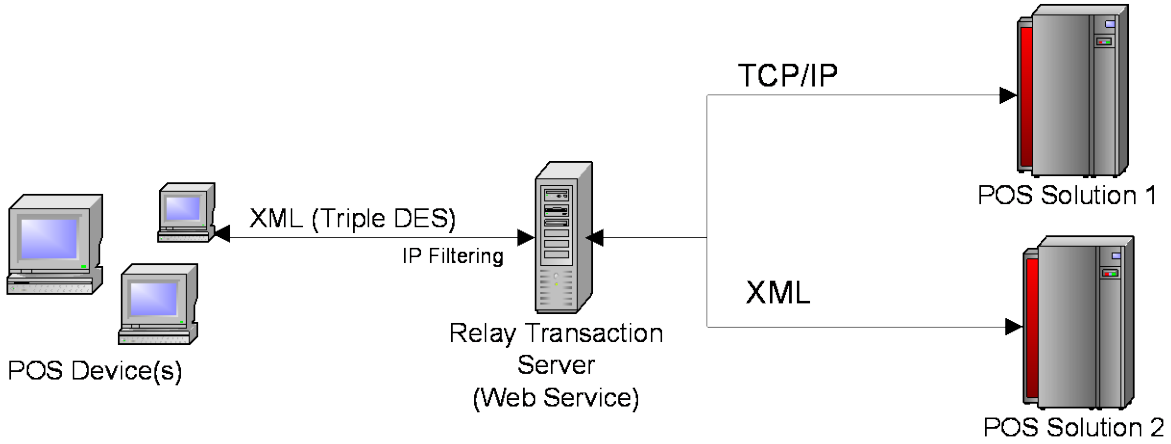
Security is a major concern for ARAMARK Campus Services. For this reason, we have implemented Triple DES encryption between the POS application and our Relay Transaction Server. Triple DES is a symmetric encryption algorithm and works using two pieces of data, a key and an initialization vector. These two items are combined and used with the data to be encrypted to create the cipher text. This cipher text can only be decrypted using the key and initialization vector used to encrypt the original data.

The Relay Transaction Server will use the appropriate encryption method for the POS System. A unique Triple DES encryption algorithm will be setup for each POS application/device accessing the Relay Transaction Server.

For security considerations, future releases of this transaction specification will result in the expiration of the Triple DES encryption key. These keys will then need to be resynchronized between the POS application/device and the Relay Transaction Server.

Transaction Specification

The transactions currently supported by our Relay Transaction Server are detailed below. These Transaction specifications are designed to interact between multiple POS system but use a standard XML schema for delivery of the information. Some information may not be available depending on the POS solution the Relay Transaction Server is processing the data though.



Inquiry Transaction Specification

The Inquiry method allows for inquires to be requested to the POS system. These requests will use ARAMARK's standard XML schema to deliver the data to the Relay Transaction Server. An XML response will be sent back with the detailed information below. Please see the XML schema definitions at the end of this document for more information. Inquiry transactions are always online, so the Offline tag must be set to FALSE.

Request/ Response	Tag	Name	Description	Encrypted	Required
Request	POSLOG	Account	Account Name ** Must be same name as setup on Relay Transaction Server & relates to a POS device.		*
Request	Transaction	BeginDatetimeStamp	Beginning Date and Time that the Transaction was Started.		*
Request	Transaction	EndDateTimeStamp	Ending Date and Time that the Transaction was Started, USE BeginDatetimeStamp if not available.		*
Request	Transaction	OperatorID	Operation Number from system sending over request.	Y	*
Request	Transaction	TransactionTypeCode	"INQUIRY"	Y	*
Request	Transaction	TransactionID	TransactionID for the inquiry statement.	Y	
Request	Transaction	RevenueCenterID	Optional Field to record transactions to a Revenue Center	Y	
Request	Transaction	OffLine	"FALSE"	Y	*
Request	Inquiry	CustomerID	Customer Account Number	Y	*
Response	Transaction	DateTimeStamp	Date & Time of the Transaction		*
Response	Transaction	TransactionTypeCode	"INQUIRY"	Y	*
Response	Transaction	Status	Status of transaction.	Y	*
Response	Transaction	StatusCode	Transaction success	Y	*
Response	Transaction	ResponseMessage	Response message from POS System	Y	
Response	Transaction	ResponseText	Response Text from POS system	Y	
Response	Customer	CustomerID	Customer Card Number	Y	*
Response	Customer	CustomerIssueID	Customer Card Number Issue Number	Y	
Response	Customer	FirstName	FirstName of Customer	Y	
Response	Customer	LastName	LastName of Customer	Y	

Response	Customer	Amount	Amount on Customer Account.	Y	*
Response	TaxLines	TaxAuthority	Number of Tax	Y	*
Response	TaxLines	UseTax	TRUE or FALSE	Y	*

Inquiry Transaction Specification Request

Below is an example of a inquiry transaction to a ScanPlus POS system.
** Encryption is not enabled for this example.

```
<POSLOG>
  <Account>DEVICE_1</Account>
   <Transaction>
        <BeginDateTimeStamp>2004-04-14T10:13:45.0000000-04:00</BeginDateTimeStamp>
        <EndDateTimeStamp>2004-04-14T10:13:45.0000000-04:00</EndDateTimeStamp>
        <OperatorID>1</OperatorID>
        <TransactionTypeCode>INQUIRY</TransactionTypeCode>
        <TransactionID>00001</TransactionID>
        <RevenueCenterID>9999</RevenueCenterID>
        <OffLine>False</OffLine>
      <Inquiry>
           <CustomerID>000000001</CustomerID>
      </Inquiry>
   </Transaction>
</POSLOG>
```

Below is an example of an inquiry response from a ScanPlus POS System.
** Encryption is not enabled for this example.

```
 <POSResponse>
    <Transaction>
        <DateTimeStamp>2004-04-14T10:13:45.0000000-04:00</DateTimeStamp>
        <TransactionTypeCode>INQUIRY</TransactionTypeCode>
        <Status>Valid</Status>
        <StatusCode>0</StatusCode>
        <ResponseMessage>Customer Balance is 500.00</ResponseMessage>
        <ResponseText />
       <Customer>
           <CustomerID>11111111</CustomerID>
           <CustomerIssueID>01</CustomerIssueID>
           <FirstName>First</FirstName>
           <LastName>Name</LastName>
           <Amount>500.00</Amount>
           <Tax>
              <Tax>
                  <TaxAuthority>1</TaxAuthority>
                  <UseTax>TRUE</UseTax>
              </Tax>
           </Tax>
        </Customer>
     </Transaction>
  </POSResponse>
```

Declining Balance Transaction Specifications

The Declining Balance transaction method allows for debits to be deducted from a customers account. These requests will use ARAMARK's standard XML schema to deliver the data to the Relay Transaction Server. An XML response will be sent back with the detailed information below. Please see the XML schema definitions at the end of this document for more information.

Request/ Response	Tag	Name	Description	Encrypted	Required
Request	POSLOG	Account	Account Name ** Must be same name as setup on Relay Transaction Server. Relates to a POS device.		*
Request	Transaction	BeginDatetimeStamp	Beginning Date and Time that the Transaction was Started.		*
Request	Transaction	EndDateTimeStamp	Ending Date and Time that the Transaction was Started, USE BeginDatetimeStamp if not available.		*
Request	Transaction	OperatorID	Operator ID for this transaction.	Y	*
Request	Transaction	TransactionTypeCode	"DECLININGBALANCE"	Y	*
Request	Transaction	TransactionID	TransactionID for the declining balance request.	Y	
Request	Transaction	RevenueCenterID	Optional Field to record transactions to a Revenue Center	Y	
Request	Transaction	OffLine	Offline = "FALSE" Online = "TRUE"	Y	*
Request	DecliningBalance	CustomerID	Customer Card Number	Y	*
Request	DecliningBalance	Amount	Amount to be deducted from the customers Account	Y	*
Response	Transaction	DateTimeStamp	Date & Time of the Transaction		*
Response	Transaction	TransactionTypeCode	"DECLININGBALANCE"	Y	*
Response	Transaction	Status	Status of transaction.	Y	*
Response	Transaction	StatusCode	Transaction success	Y	*
Response	Transaction	ResponseMessage	Response message from POS System	Y	
Response	Transaction	ResponseText	Response Text from POS system	Y	
Response	Customer	CustomerID	Customer Card Number	Y	*
Response	Customer	CustomerIssueID	Customer Card Number Issue Number	Y	
Response	Customer	FirstName	FirstName of Customer	Y	
Response	Customer	LastName	LastName of Customer	Y	
Response	Customer	Amount	New Balance on Student Acount.	Y	*

Recommend Practices:

These practices are recommended so that you can verify the transactions have been successfully completed in your application.

Campus Services IT recommends that prior to a declining balance transaction, you first perform a Inquiry Transaction to verify the student's balance. This will initially verify that your connection to ScanPlus is online.

Below is an example of a Declining Balance request to a ScanPlus POS System.
** Encryption is not enabled for this example.

```
<POSLOG>
  <Account>DEVICE_1</Account>
  <Transaction>
    <BeginDateTimeStamp>2004-04-14T10:13:45.0000000-04:00</BeginDateTimeStamp>
    <EndDateTimeStamp>2004-04-14T10:13:45.0000000-04:00</EndDateTimeStamp>
    <OperatorID>1</OperatorID>
    <TransactionTypeCode>DECLININGBALANCE</TransactionTypeCode>
    <TransactionID>00001</TransactionID>
    <RevenueCenterID>9999</RevenueCenterID>
    <OffLine>False</OffLine>
```

```
      <DecliningBalance>
          <CustomerID>000000001</CustomerID>
          <Amount>5.60</Amount >
        </DecliningBalance>
    </Transaction>
  </POSLOG>
```


Below is an example of a Declining Balance response from a ScanPlus POS System.
** Encryption is not enabled for this example.


```
<POSResponse>
  <Transaction>
      <DateTimeStamp>2004-04-14T10:13:45.0000000-04:00</DateTimeStamp>
      <TransactionTypeCode>DECLININGBALANCE</TransactionTypeCode>
      <Status>Valid</Status>
      <StatusCode>0</StatusCode>
      <ResponseMessage>Customer Balance is 500.00</ResponseMessage>
      <ResponseText />
     <Customer>
        <CustomerID>11111111</CustomerID>
        <CustomerIssueID>01</CustomerIssueID>
        <FirstName>First</FirstName>
        <LastName>Name</LastName>
        <Amount>500.00</Amount>
      </Customer>
    </Transaction>
  </POSResponse>
```

Credit Transaction Specifications

The Credit transaction method allows for credits/refunds to be placed onto a customers account.. An XML response will be send back with the detailed information below. In validating the data, a check should first be made to the Status field to verify that the transaction was successful.

Request/ Response	Tag	Name	Description	Encrypted	Required
Request	POSLOG	Account	Account Name ** Must be same name as setup on Relay Transaction Server. Relates to a POS device.		*
Request	Transaction	BeginDatetimeStamp	Beginning Date and Time that the Transaction was Started.		*
Request	Transaction	EndDateTimeStamp	Ending Date and Time that the Transaction was Started, USE BeginDatetimeStamp if not available.		*
Request	Transaction	OperatorID	Operator ID for this transaction.	Y	*
Request	Transaction	TransactionTypeCode	"CREDIT"	Y	*
Request	Transaction	TransactionID	TransactionID for the credit transaction.	Y	
Request	Transaction	RevenueCenterID	Optional Field to record transactions to a Revenue Center	Y	
Request	Transaction	OffLine	Offline = "FALSE" Online = "TRUE"	Y	*
Request	Credit	CustomerID	Customer Card Number	Y	*
Request	Credit	Amount	Amount to be Credited to the customers account.	Y	*
Response	Transaction	DateTimeStamp	Date & Time of the Transaction		*
Response	Transaction	TransactionTypeCode	"CREDIT"	Y	*
Response	Transaction	Status	Status of transaction.	Y	*
Response	Transaction	StatusCode	Transaction success	Y	*
Response	Transaction	ResponseMessage	Response message from POS System	Y	
Response	Transaction	ResponseText	Response Text from POS system	Y	
Response	Customer	CustomerID	Customer Card Number	Y	
Response	Customer	CustomerIssueID	Customer Card Number Issue Number	Y	*
Response	Customer	FirstName	FirstName of Customer	Y	
Response	Customer	LastName	LastName of Customer	Y	
Response	Customer	Amount	New Balance on Student Acount.	Y	

Recommend Practices:

These practices are recommended so that you can verify the transactions have been successfully completed in your application.

Campus Services IT recommends that prior to a credit transaction, you first perform a inquiry transaction to verify the student's balance.

Below is an example of a Credit request to a ScanPlus POS System.
** Encryption is not enabled for this example.

```
<POSLOG>
   <Account>DEVICE_1</Account>
  <Transaction>
      <BeginDateTimeStamp>2004-04-14T10:13:45.0000000-04:00</BeginDateTimeStamp>
      <EndDateTimeStamp>2004-04-14T10:13:45.0000000-04:00</EndDateTimeStamp>
      <OperatorID>1</OperatorID>
      <TransactionTypeCode>CREDIT</TransactionTypeCode>
      <TransactionID>00001</TransactionID>
      <RevenueCenterID>9999</RevenueCenterID>
      <OffLine>False</OffLine>
     <Credit>
```

```
        <CustomerID>000000001</CustomerID>
        <Amount>5.00</Amount>
      </Credit>
    </Transaction>
  </POSLOG>
```


Below is an example of a Credit response from a ScanPlus POS System.
** Encryption is not enabled for this example.


```
<POSResponse>
  <Transaction>
      <DateTimeStamp>2004-04-14T10:13:45.0000000-04:00</DateTimeStamp>
      <TransactionTypeCode>CREDIT</TransactionTypeCode>
      <Status>Valid</Status>
      <StatusCode>0</StatusCode>
      <ResponseMessage>Customer Balance is 500.00</ResponseMessage>
      <ResponseText />
     <Customer>
        <CustomerID>11111111</CustomerID>
        <CustomerIssueID>01</CustomerIssueID>
        <FirstName>First</FirstName>
        <LastName>Name</LastName>
        <Amount>500.00</Amount>
      </Customer>
    </Transaction>
  </POSResponse>
```

ARAMARK – POS Interfaces

The following Schema is defined to assist third party venders in the development of transacting between the ARAMARK Campus Services Relay Transaction Server and the POS system.

Element	Definition
POSLog	The record of all activity that is performed at a POS by an operator.

POSLog

Tag	Type	Definition
POSLog	struct	Root element of POSLog.
Account	string	ARAMARK Relay Transaction server code to determine which POS Solution to relay the transaction to.
Transaction	struct	A single Transaction that's arrived from a POS.

Transaction

Tag	Type	Definition
Transaction	struct	A single Transaction that's arrived from a POS.
BeginDateTimestamp	string	The time and date a transaction is initiated.
EndDateTimestamp	string	The time and date a transaction is completed.
* OperatorID	string	A unique, automatically assigned number used to identify a workstation OPERATOR.
* TransactionTypeCode	string	A code to denote the type of TRANSACTION, eg RETAIL, BOARD, INQUIRY, DECLININGBALANCE, CASHEQUIVALENCY, CREDIT
* TransactionID	string	A Unique Identifier for a transaction. * Restrictions apply for each POS System.
* RevenueCenterID	string	Identifier for the RevenueCenter to which this transaction is to be credited.
Retail Transaction	struct	A subtype of transaction that records the sale & return of merchandise & services with customers.
Board	struct	A subtype of transaction that records the use of a Board transaction with customers. (non-monetary)
Inquiry	struct	A subtype of transaction that requests an Inquiry of a customer.
Declining Balance	struct	A subtype of transaction that records the use of a Declining Balance amount with a customer.
Cash Equivalency	struct	A subtype of transaction that requests the Cash Equivalency for a customer.
Credit	struct	A subtype of transaction that requests a Credit to put onto a customers account.
Offline	String	Value to determine if the Tender is Offline on a transaction TRUE, FALSE

Declining Balance

Tag	Type	Definition
BecliningBalance	struct	A subtype of transaction that deducts a value from an account for a customer in a value stored account.
* CustomerID	string	A unique identifier for the customer that conducted the Declining Balance.
* Amount	string	The total amount to be deducted from the store valued customer account.

Credit

Tag	Type	Definition
Credit	struct	A subtype of transaction that records a Credit to a customers account.
* CustomerID	string	A unique identifier for the customer.

* Amount	string	The total amount to be credited to the customers account.

Inquiry

Tag	Type	Definition
Inquiry	struct	A subtype of transaction that records the request of an inquiry for a particular customer.
* CustomerID	string	A unique identifier for the customer that conducted the Inquiry.

POS Responses

The following Schema RootElements are defined in the XML Dictionary to assist in navigation.

Element	Definition
POSResponse	POS Responses

POSResponse

Tag	Type	Definition
POSresponse	struct	Root element of POSResponse.
Transaction	struct	A single Transaction that's been responded to from a POS.

Transaction

Tag	Type	Definition
Transaction	struct	Subtype of POSResponse
DateTimeStamp	String	A single Transaction that's been responded to from a POS.
TransactionTypeCode	string	INQUIRY, DECLININGBALANCE, CASHEQUIVALENCY, BOARD
* Status	string	Transaction success code (Valid, InValid, UnKnown, NotRegistered)
* StatusCode	string	POS Transaction response code
* ResponseMessage	string	POS message, Potentially for Customer.
* ResponseText	string	POS Text / Error Messages
* Customer	string	Subtype of Transaction that this transaction relates to

Customer

Tag	Type	Definition
Customer	struct	Subtype of Transaction
* CustomerID	String	Customer Card Number
* CustomerIssueID	String	Customer Card Number Issue Number
* FirstName	String	Customer First name
* LastName	String	Customer Last name
* Amount	String	Customer Account Balance
*TaxLines	struct	Structure of TaxLines

TaxLines

Tag	Type	Definition
TaxAuthorityID	String	TaxNumber
UseTax	String	TRUE or FALSE value if tax is used.